Experiment 6: Time Series Analysis of Website Traffic

Aim :

To analyze the time series data of website traffic to identify trends, seasonality, and anomalies.

Program Code :

```
import pandas as pd
import matplotlib.pyplot as plt
from statsmodels.tsa.seasonal import seasonal_decompose
from google.colab import files

uploaded = files.upload()
# Load dataset
data = pd.read_csv(
    'website_traffic.csv',
    parse_dates=['Date'],
    index_col='Date'
)

# Time Series Decomposition
decomposition = seasonal_decompose(
    data['Page_Views'],
    model='additive',
    period=7
)

decomposition.plot()
plt.show()

# Moving Average
data['Page_Views_MA'] = data['Page_Views'].rolling(window=30).mean()
```

```
data[['Page_Views', 'Page_Views_MA']].plot(
    figsize=(10, 5)
)

plt.title('Page Views with Moving Average')
plt.xlabel('Date')
plt.ylabel('Page Views')
plt.show()

# Anomaly Detection using Z-score
data['Page_Views_Z'] = (
    (data['Page_Views'] - data['Page_Views'].mean())
    / data['Page_Views'].std()
)

# Find anomalies
anomalies = data[data['Page_Views_Z'].abs() > 3]

# Plot anomalies
plt.figure(figsize=(10, 5))

plt.plot(
    data.index,
    data['Page_Views'],
    label='Page Views'
)

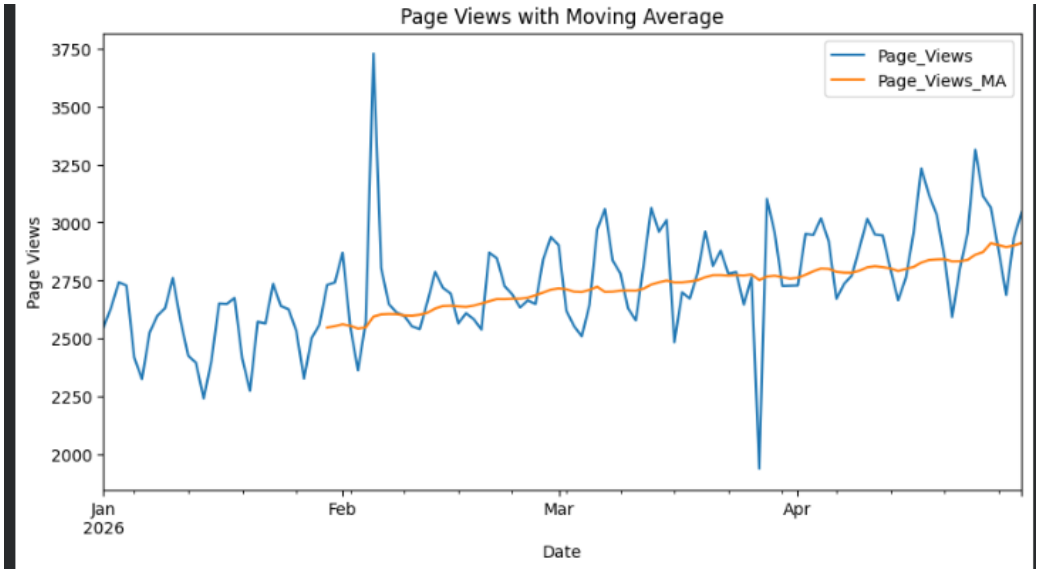
plt.scatter(
    anomalies.index,
    anomalies['Page_Views'],
    color='red',
```

```
    label='Anomalies'
)

plt.title('Page Views with Anomalies')
plt.xlabel('Date')
plt.ylabel('Page Views')
plt.legend()
plt.show()

# Display anomalies
print("Anomalies:")
print(anomalies[['Page_Views', 'Page_Views_Z']])
```

Output :



Result :

Thus, Time Series Analysis of Website Traffic were Programmed.